

random dither, by only dithering when it recognises a shift in the load point.

Combining the assorted switchers into fewer direct-conversion switchers will save both switching and FR losses in the system, while actually decreasing the complexity of the system at the same time. It will, however, increase the need for intelligent coordination between the various systems. The MPP will have to communicate with the battery charger logic to coordinate the energy transfer to the batteries, and the MPP will have to coordinate with the battery-boost regulators to effectively manage the multiple bus voltages for the load. As a result, the final system will require some coordinating intelligence to manage the ebb and flow of energy through the system.

The power savings of a more intelligent dithering algorithm, as mentioned above, is two fold. One, finding the maximum power point faster limits inefficient energy transfer losses during the search. Two, making a change only when a change is detected will limit the waste, due to the periodic inefficient test dithers. This also allows the MPP to respond immediately to a change, rather than incurring a loss while waiting for the periodic dither.

The new dither does require that the MPP have the ability to detect a change in the solar cell outputs. To accomplish this, the new intelligent MPP only needs to watch the output voltage of the solar cells. The solar cells actually put out the same voltage, over most lighting conditions—only their source resistance changes with the light level. So, the voltage at the output of the solar cells will only change if either the load changes or the amount of available light changes. That means, if the voltage at the output of the solar cell goes up without a change in load, then there is more energy available and the MPP should start increasing the load until it reaches the new maximum power point. If the voltage drops, then the amount of available current has dropped and the MPP should decrease the load until the lower maximum

power point is reached. If the voltage doesn't change, the MPP doesn't need to do anything. It can just sleep in a low current mode until a voltage change is detected. Furthermore, by measuring the shift in solar-cell voltage, the MPP can even approximate the amount of shift in the maximum power point and jump to a close approximation—before dithering into the optimum value. This saves time and energy.

There is one last optimisation to the system, although it is not really a power system change. We can make our load intelligent. Remember that the MPP is trying to pull every milliwatt of power out of our solar cells. Most of this is to get the maximum amount of energy from the solar system. However, it is also to keep the solar cells cooler. By making the load intelligent, we can schedule high current usages during times when there is an excess of energy available from the solar cells (full sun and fully charged batteries). Then, no energy is wasted as heat in either the solar cells or the battery. Instead, it can be routed to the load to do useful work. Being intelligent, the load can also scale back its requirements when there is less energy available—reducing losses in the battery boost circuits. Basically, by intelligently scheduling the work done by the load, the load makes the most of the available energy in the system. And, it does so at times when the best conversion efficiency is available, rather than pulling power from the storage capability with the associated higher efficiency losses.

To summarise, if we want to create an optimally efficient system, we need the following:

- An optimised mix of solar cells, in strings, with an optimum number of MPP converters to minimize switching, voltage-related, and current-related losses in the MPP converters.
- An optimised mix of battery cells stacked in strings, with individual chargers and output boost switchers. The optimal mix being the configuration that minimizes switching,

voltage-related, and current-related losses in the charges and output boost switchers, while limiting the impact of a potential cell failure.

- A hybrid of charger logic and MPP current output switching, to create an efficient charging path between the solar cells and the batteries.
- A hybrid load-share combination of battery-boost switcher and MPP converter to directly generate the supply voltages for the load with a minimum loss.
- An intelligent MPP converter dithering algorithm that only changes loading when needed, using a fast detection and search algorithm in place of a periodic random dither.
- An intelligent load that is cognizant of the power system status, so it can take advantage of higher efficiency periods for higher current tasks.
- An implied central control intelligence, charged with coordinating the power conversion between solar, battery and load.

Taking an organised approach to a solar power system can significantly improve the overall efficiency of the system. However, it does so by increasing the complexity of the system. In addition, it relies heavily on intelligent control to achieve many of the gains in efficiency and reliability. Fortunately, several small microcontrollers introduced in recent years have on-chip mixed-signal peripherals that are capable of implementing most, if not all, of the conversion functions described here. So, while the complexity may have increased, it is likely that the actual number of chips required for the design is reduced—due to the consolidation of switching functions into the microcontrollers. This only goes to show that adding a little intelligence to a design is typically a better option than using a brute-force approach—regardless of how simple the brute-force approach may seem. ●

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